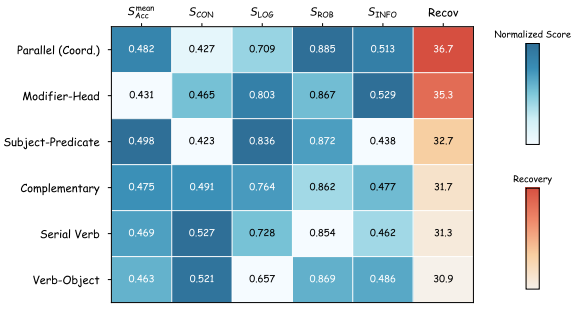

Figure 5: **Performance by syntactic category.** Parallel structures are easiest due to redundancy and symmetry cues, while asymmetric dependency patterns remain harder for reconstruction. Metrics are reported to diagnose whether failures arise from semantic gaps, instability, or structural sensitivity.

their canonical neighborhoods yield higher recovery, while dispersed anchors significantly degrade performance.

Few-shot prompting yields the most uniform improvements across permutations, suggesting that explicit exemplars constrain decoding more effectively than generic reasoning traces. In contrast, CoT gains are less stable and may amplify format drift in weaker models.

4.4 Impact of Script Typology

Table 4 compares performance across the four script settings. We observe that reconstruction difficulty is strongly shaped by script typology.

Logographic scripts provide stronger local anchors. Chinese and Japanese exhibit substantially higher recovery than Korean. Logographic characters preserve standalone semantic cues under scrambling, providing local anchors that facilitate global reassembly.

Hangul scrambling is a strict negative control. Korean Hangul behaves differently: internal scrambling destroys syllable composition and removes meaningful local cues. As a result, recovery collapses sharply, confirming that reconstruction depends not only on positional decoding but also on the preservation of interpretable sub-units.

Consistency inflation under extreme ambiguity. Notably, Korean yields high consistency despite low recovery. Under severe uncertainty, models often default to generic, stable explanations across permutations. This yields artificially high invariance but does not reflect true reconstruction ability, reinforcing the need to interpret diagnostic metrics jointly.

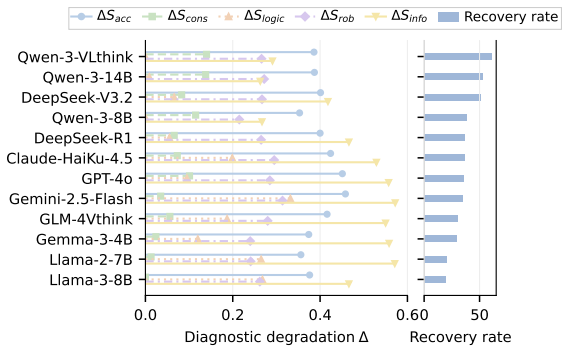

Figure 6: **Degradation under perturbed 3-shot inputs.** Left: per-metric drops ΔS_{acc}, ΔS_{cons}, ΔS_{logic}, ΔS_{rob}, ΔS_{info}, computed as *unperturbed* minus *perturbed 3-shot* under matched inference settings. Right: recovery rate for perturbed 3-shot inputs. Models are ordered by recovery rate (higher is more robust).

4.5 Analysis of Syntactic Patterns

Figure 5 reports results across six syntactic categories, revealing systematic structural variation beyond surface shuffling.

Parallel structures are easiest. Parallel and coordinate patterns are generally easier due to redundancy and symmetric templates. These cues effectively reduce the search space and provide ordering heuristics.

Meaning does not imply correct ordering. For subject–predicate patterns, models often recover the underlying proposition but fail to linearize characters into the precise canonical order, confirming that semantic grasp does not guarantee structural planning.

Asymmetric dependencies remain difficult. Verb–object patterns are harder because they rely on directional dependencies that are disrupted by permutation noise, leading to semantically plausible but structurally incorrect guesses.

4.6 Analysis of In-Context Learning

Figure 6 reports results under three-shot prompting.

Perturbations reveal structured degradation rather than uniform noise. Across models, diagnostic drops concentrate on a subset of components instead of moving in lockstep, forming consistent "degradation fingerprints" per model. This suggests perturbations disrupt specific capabilities (e.g., information grounding or robust constraint use) rather than inducing indiscriminate instability, consistent with the separation between fine-grained deltas and the overall recovery rate in the figure.

Input (Scrambled): 城 (City), 倾 (Topple), 国 (State), 倾 (Topple)

Target: 倾国倾城(Qing Guo Qing Cheng). Meaning: Extreme female beauty.

Qwen-3-8B **Literal Hallucination**

Generated explanation:
指国家和城池被攻破或毁灭，常用于形容重大失败或灾难性后果。*(Refers to a country or city being breached.)*

Error note: Structurally fluent output (high S_{Cons}) but incorrect logic due to literal character-level composition.

DeepSeek-V3 **Precise Retrieval**

Generated explanation:
原指君主过分迷恋女色而亡国。后形容女子容貌极美。*(Originally refers to a monarch losing his kingdom.)*

Error note: Correctly captures the idiomatic metaphorical meaning.

Figure 7: **Case Study on Semantic Hallucination.** Qwen-3-8B produces a fluent but incorrect literal interpretation, while DeepSeek-V3 correctly retrieves the idiomatic metaphor.

High recovery does not imply uniformly preserved diagnostics. Several models sustain relatively strong recovery while still exhibiting pronounced drops on particular diagnostics, indicating compensatory behavior: outputs can be reconstructed at the sequence level even when intermediate structural signals degrade. Conversely, models with low recovery typically show broader declines across multiple diagnostics, suggesting recovery failures coincide with multi-faceted structural breakdown rather than a single isolated weakness.

4.7 Case Study: Semantic Hallucination Induced by Literal Compositionality

To illustrate how semantic plausibility can mask reconstruction failure, Figure 7 presents a representative case on *Qing Guo Qing Cheng*. We observe two contrasting behaviors: some models retrieve the idiom-level metaphor correctly, while others interpret characters literally and hallucinate a compositional meaning (e.g., "toppling a country and a city"). This example highlights why OrderProbe requires deterministic reconstruction and diagnostic metrics beyond surface semantic similarity.

4.8 Memorization and Frequency Analysis

A natural concern is whether reconstruction mainly reflects memorization of frequent expressions. We therefore conduct a pilot frequency-stratified analysis (Appendix B.4). Although recovery improves with idiom frequency, reconstruction failures remain common even in high-frequency bins, suggesting that OrderProbe captures structural limitations beyond pure coverage effects.

5 Related Work

Robustness and Adversarial Evaluation The evaluation of model robustness has evolved from surface-level perturbations to semantic integrity. Early benchmarks like TextFlint (Wang et al., 2021) and CheckList (Ribeiro et al., 2020) primarily focused on lexical noise, such as typos and synonym substitution. More recent works have examined robustness in instruction following (Li et al., 2024) and logical reasoning (Lin et al., 2025). Unlike general adversarial attacks that seek to break model performance with noise, OrderProbe probes the cognitive limit of reconstructing canonical order from semantically valid but positionally disordered inputs.

Idiom and Figurative Language Processing Prior research on idioms has predominantly treated them as static lexical units. Datasets such as ChID (Zheng et al., 2019) and Chengyu-Bench (Fu et al., 2025) focus on cloze tests, retrieval, or translation (Donthi et al., 2025). While these benchmarks evaluate whether a model knows an idiom, they do not test whether the model understands its composition. As demonstrated in our experiments, models often exhibit a "knowledge-structure dissociation," performing well on retrieval tasks while failing on the reconstruction tasks proposed in OrderProbe.

Positional Bias and Structural Reasoning Our findings resonate with theoretical studies on the positional limitations of Transformer architectures. Sinha et al. (2021) observed that models are often insensitive to word order in classification tasks, while Berglund et al. (2024) identified the "Reversal Curse," highlighting the inability of models to generalize relationships in reverse directions. OrderProbe provides empirical evidence for these phenomena in the context of East Asian linguistic structures, confirming that even high-resource languages like Chinese are susceptible to severe structural fragility when local semantic anchors are displaced.

6 Conclusion

We introduce **OrderProbe**, a deterministic benchmark for structural reconstruction using four-character expressions in Chinese, Japanese, and Korean. Experiments on twelve LLMs show that

exact recovery remains difficult, with zero-shot performance often below 35%, revealing a clear semantic–structure dissociation. We further find that reconstruction success depends strongly on script typology, where logographic scripts provide stronger local anchors and Hangul serves as a strict negative control. We release OrderProbe and our diagnostic framework to support future research on structural robustness in LLMs.

Limitations

Our OrderProbe is designed to evaluate model capabilities through structural reasoning tasks. The types within our dataset reflect our effort to collect data from multiple perspectives. However, we acknowledge that the scope of knowledge covered is inherently limited by the expertise of the contributors and cannot encompass all facets of linguistic structural integrity. Moreover, structural robustness currently lacks a universally accepted definition. Our evaluation is based on six syntactic categories that, while comprehensive, do not cover the full spectrum of linguistic patterns.

Since our dataset is constructed by referencing materials in three languages, we provide multilingual evaluation data. However, the influence of different script typologies on model performance has not been thoroughly analyzed for alphabetic languages. We have not conducted detailed experiments on morphologically rich systems where structural perturbations might yield different results.

In addition, some data points involve crystallized expressions that are present in pre training corpora. We recognize that correct reconstruction depends on the memorization of these fixed patterns. Although the results are accurate within this specific domain, future developments may require models to handle novel combinations beyond memorized content. For these items, we ensure their validity within the current linguistic context. Although these limitations are relevant, we believe they extend beyond the scope and original intent of our work. We hope that future studies will address these challenges in more detail.

Ethics Statement

The OrderProbe dataset integrates linguistic resources from the public domain under fair use principles for academic research. We confirm that the collected expressions are free from copyright restrictions and strictly adhere to data usage policies. All participants provided informed consent and followed a rigorous protocol to exclude offensive or discriminatory content.

Our data construction pipeline utilizes LLMs for semantic augmentation in full compliance with provider terms of service. We declare that all generated outputs underwent human verification to ensure factual accuracy and safety. This work serves as a diagnostic benchmark for enhancing model robustness. We foresee no direct negative societal impacts or misuse risks associated with the release of this dataset.

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